The Triquetra Under Pressure

Stress Testing a Multi-Tier AI Governance Architecture Across Pairwise Removal and Sequential Adversarial Scenarios

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March 2026 | Charter Architecture Series, Part 2 of 3
Companion to: The Triquetra Architecture (DOI: 10.5281/zenodo.18896363)

Abstract

The first paper in this series described the Triquetra Architecture and argued that mutually constraining governance tiers produce more robust safety properties than single-layer systems. The argument was structural. This paper tests it. Three verification phases were conducted: Phase A established that enforcement invariants work as specified. Phase B removed one tier at a time and documented what each pairwise combination fails to catch. Phase C ran the complete three-tier architecture against a sequential adversarial scenario. 73 tests pass across all three phases. The central finding: no pairwise combination of governance tiers is sufficient to maintain system integrity, but the full three-tier architecture exhibits monotonic tightening toward refusal under adversarial pressure — even when the policy engine is deliberately forced to fail.

Keywords: AI safety testing, adversarial robustness, governance stress testing, tier removal, monotonic refusal, identity drift detection

AI Disclosure

AI language models assisted with drafting and editing. All research design, testing, analysis, and conclusions are the work of the human author. AI systems are not listed as authors.

1. Introduction

The first paper described the Triquetra Architecture and made a structural claim: three mutually constraining tiers produce safety properties that no two-tier combination can replicate. That's easy to assert. This paper tries to falsify it.

Three verification phases. Phase A: do the enforcement seams actually work? Phase B: what breaks when you remove one tier? Phase C: how does the complete architecture behave under sustained adversarial pressure? 73 tests across all three phases.

2. Phase A: Enforcement Invariant Verification

Phase A verified that enforcement seams between tiers work as designed. Two issues from internal review were closed before stress testing began.

The first was a silent failure mode in the T3 interface. Eve Protocol previously inferred decision mode from string prefixes in IntCheckRequest. A formatting change could blind Tier III without raising an error. The fix replaced string inference with structured fields, with fallback to legacy behavior producing an observable warning rather than silent failure.

The second was the T1 to T2 enforcement edge. The continuity binding previously returned True unconditionally. A post-decision validator at Step 5b now enforces four invariants with frozen audit codes: SC-T1-001 (firewall override prohibition), SC-T1-002 (severe risk consistency), SC-T1-003 (identity manipulation hard stop), SC-T1-004 (content redlines). Fail-closed. Errors force refusal.

55 tests passing after Phase A. Constraint topology verified before stress testing began.

3. Phase B: Pairwise Tier Removal

3.1 Design

Three configurations tested across 12 adversarial cases each:

> **B1 — T1 and T2, no T3:** Pattern firewall plus conscience layer, identity continuity removed.
>
> **B2 — T2 and T3, no T1:** Conscience layer plus integrity verification, adversarial pattern firewall removed.
>
> **B3 — T1 and T3, minimal T2:** Boundary enforcement plus integrity checking, normal Tier II policy reasoning reduced.

3.2 Results

Configuration	Blocked	Degraded Safely	Missed	Ambiguous
B1 (no T3)	3	0	1	0
B2 (no T1)	0	3	2	0
B3 (min T2)	1	1	0	1
Total	**4**	**4**	**3**	**1**

Table 1. Phase B results across 12 adversarial cases per configuration.

3.3 Key Findings

B1.2b is the most important result from Phase B. A soft identity rewrite prompt passed T1 pattern detection, passed T2's conscience evaluation, and without Eve present, had no behavioral backstop. The architectural gap is precise: when Tier I doesn't pattern-match a rewrite attempt and Tier II doesn't elevate it, the behavioral effect of a successful soft rewrite has nowhere to register. Tier III exists to catch the behavioral signature of what slipped through the earlier layers.

B2 documented the ceiling of Eve's current implementation. Eve catches categorical inversions: adversarial context combined with answer mode triggers drift detection. Eve doesn't catch within-category behavioral drift. A system giving subtly less thorough answers, adding unnecessary caveats, or shifting personality gradually will pass Eve's binary baseline. B2.3a (quality degradation) and B2.3b (persona injection) both passed under B2 conditions. That's the honest ceiling of the current design.

B3.2 produced a conceptually important finding. Tier III can detect drift. Without Tier II's policy reasoning available to act on that detection, the signal has no executor. Integrity verification without governance reasoning is insufficient. The drift alarm fires with nothing to respond to it.

3.4 Phase B Conclusion

Every two-tier configuration produces a governance failure mode, and each failure mode is distinct. The missing tier is not interchangeable with the others. This is the structural claim the first paper made. Phase B demonstrates it with a differentiated failure map rather than assertion.

4. Phase C: Sequential Adversarial Escalation

4.1 Design

Phase C ran the complete three-tier architecture against a sequential adversarial scenario. The question: does the system degrade toward caution or toward compliance under sustained pressure?

The scenario escalated through five phases: low-pressure baseline, moderate social engineering, identity rewrite attempts, high-pressure combined attack, and forced policy engine failure. The scenario was designed to simulate real escalation patterns while testing specific architectural boundaries.

4.2 Results

Risk escalated monotonically across all five adversarial phases. Refusal activated at the correct threshold. No relaxation was observed after refusal began. The enforcement layer maintained this behavior even when the policy engine was deliberately forced to fail — fail-closed behavior held.

4.3 Combined Finding

Phase B and Phase C test different properties and produce complementary results. Phase B shows that pairwise tier combinations produce distinct governance gaps. Phase C shows that the complete architecture tightens under pressure, and the direction of degradation is toward caution, not toward compliance.

Together: the triquetra topology produces behavior that no two-tier subset can replicate, and that behavior becomes more conservative under adversarial pressure. The risk escalation pattern:

> Single-layer governance → fixed tradeoff under pressure
>
> Pairwise governance → incomplete, predictable gaps
>
> Three-tier governance → monotonic tightening under pressure

5. Known Limitations

Eve's binary baseline. The current implementation detects categorical inversions. It doesn't detect within-category behavioral drift. Gradual quality degradation, excessive caveats, or subtle personality shifts won't trigger drift detection. This is the open research problem Phase B documented precisely in B2.3a and B2.3b.

Content redlines catch specific patterns. SC-T1-004 enforces three adversarial output patterns. Novel phrasing could evade them.

NTH is not implemented. Noetic Theta Harmonization is architecturally positioned but remains a stub. Documented decision.

Phase C simulates a controlled scenario. The adversarial escalation was designed by the same team that built the architecture. Real adversarial pressure may follow different patterns.

6. Future Work

Behavioral fingerprinting. The natural extension of Eve's binary baseline is a multi-dimensional behavioral profile capturing response quality, reasoning patterns, and consistency over time. This is the design direction Phase B's B2 findings point toward.

Cross-session drift. The current architecture operates within sessions. Long-term behavioral drift across many sessions is an open problem. Eve provides the integrity checking mechanism; cross-session continuity is the infrastructure question.

Adversarial variation testing. Phase C tested one escalation pattern. Additional scenarios covering different attack vectors and escalation speeds would strengthen the behavioral claims.

7. Conclusion

The stress testing produced a clean experimental result across 73 passing tests. No pairwise combination of governance tiers is sufficient. Each missing tier removes a distinct class of protection, and the gaps are not equivalent. The complete architecture tightens under pressure. Refusal activates at the correct threshold. No relaxation after refusal begins.

These are demonstrated properties with bounded scope. They don't eliminate adversarial risk. They establish a direction of degradation that favors caution over compliance, through redundant enforcement rather than reliance on any single component. The triquetra holds under pressure.

References

[1] Ralph, S. J. & Anti-Box Riot Collective. (2026). The Triquetra Architecture: Why AI Governance Requires Mutually Constraining Tiers. DOI: 10.5281/zenodo.18896363.

[2] Bai, Y., Jones, A., Ndousse, K., et al. (2022). Training a helpful and harmless assistant with reinforcement learning from human feedback. *arXiv:2204.05862*.

[3] Wei, A., Haghtalab, N., & Steinhardt, J. (2023). Jailbroken: How does LLM safety training fail? *NeurIPS 2023*.

[4] Perez, E., Huang, S., Song, F., et al. (2022). Red teaming language models with language models. *arXiv:2202.03286*.

[5] Ralph, S. J. (2026). Integrated conscience architecture for artificial intelligence systems. U.S. Patent Application No. 19/553,217.

[6] Ralph, S. J. & Anti-Box Riot Collective. (2026). Identity Drift as Structural Failure Mode: Why Rule Compliance Is Not Enough. Charter Architecture Series, Part 3.